

Scent Capture Device (BME688 Edition)

Same goal, commercial-grade sensor stack: 10× lower BOM, full data ownership, massive developer ecosystem

1. Executive Summary

This document specifies an alternative hardware stack for the scent capture device, built around the Bosch Sensortec BME688 (or its successor BME690) instead of the SmartNanotubes smell.Board. Everything downstream — airflow architecture, pump, compute, enclosure philosophy — is identical to the original spec. Only the sensing element changes.

Why this version exists:

- The SmartNanotubes smell.Board costs €389 and ties you to their cloud-based Smell Annotator workflow.
- The Bosch BME690 8× shuttle board costs ~\$25 and ships with an on-device ML tool (BME AI-Studio) that runs on your own machine.
- Bosch sensors are in millions of phones, smart home devices, and air quality monitors — the supply chain, tooling, and community are mature.
- For a capture-and-retrieve scent library, the BME688's discrimination capability is very likely sufficient. We should validate this with the comparison protocol in Section 10 before committing.

At a glance

- Prototype BOM: approximately \$230–280 (vs. \$830+ for the smell.Board version).
 - External envelope: ~101 × 71 × 31 mm (roughly credit-card footprint, matchbox thickness).
 - Capture cycle: 5–10 seconds per sample.
 - Fingerprint dimensionality: up to 80 virtual channels (8 sensors × 10 heater steps).
 - No vendor cloud dependency — all data and models live on your infrastructure.
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2. The Key Design Change: Heater-Cycled Virtual Channels

The BME688 and its newer sibling BME690 are MEMS gas sensors — microscopic heated metal-oxide plates on a silicon chip. Each chip has ONE physical gas-sensing element. On its own, that would give you a single-number reading every second, which isn't enough to discriminate complex smells.

Bosch solved this with a clever trick called "heater cycling." The sensor's hot plate is programmed to step through a sequence of temperatures — for example, 200 °C, 250 °C, 300 °C, 350 °C, 400 °C, and back down —

and gas resistance is measured at each step. Different volatile compounds react to the hot plate differently at different temperatures, so each temperature step effectively becomes a distinct "virtual" sensor channel.

One BME688 running a 10-step heater profile behaves like 10 differently-tuned gas sensors. The 8-sensor shuttle board running the same profile in parallel gives you 80 virtual channels — or 10 channels with 8-fold noise averaging, depending on how you configure it. This is the direct equivalent of the SmartNanotubes "64-channel" architecture, just arrived at through a different physical mechanism.

3. System Architecture

The airflow topology is identical to the smell.Board version: air is pulled (not blown) through a sealed chamber, across the sensor array, past a flow verifier, through the diaphragm pump, and out the exhaust. This is non-negotiable for reproducibility.

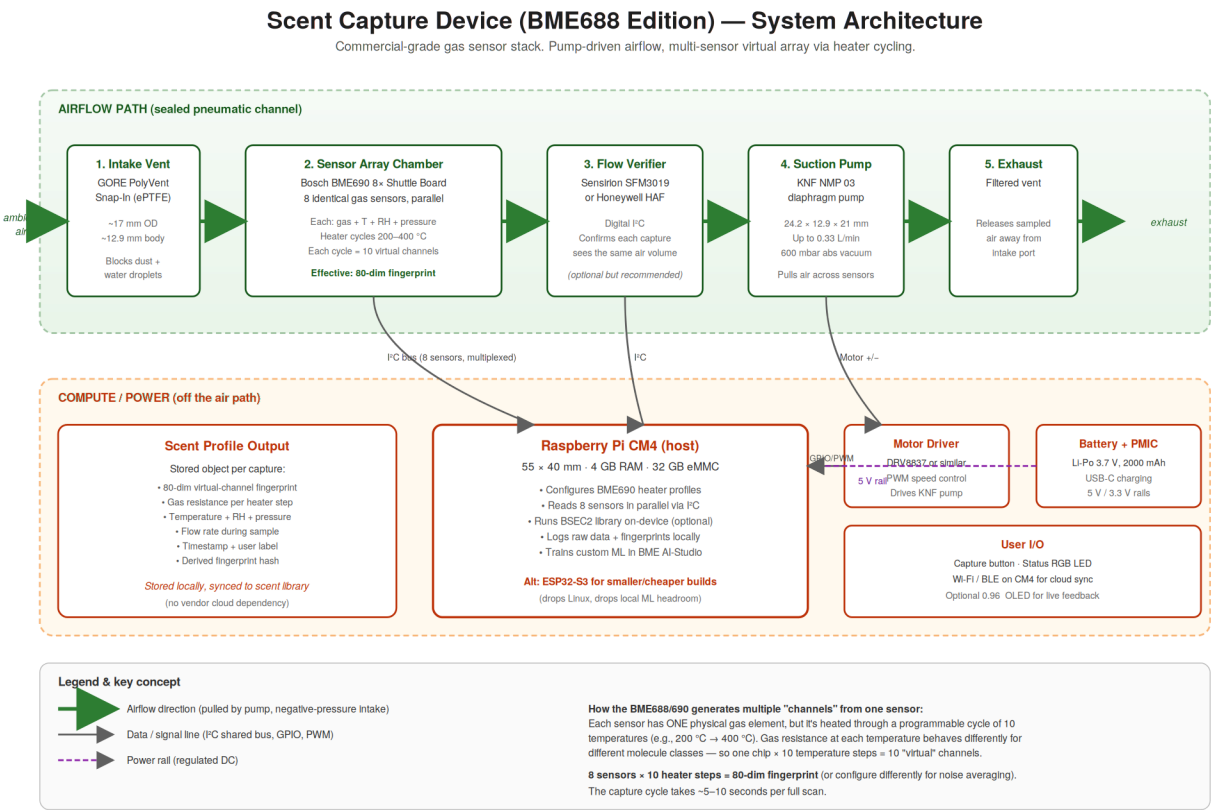


Figure 1 — BME688-edition system architecture. Green = airflow path, gray = signal lines, purple dashed = power rail.

4. Component Specifications

4.1 Primary Sensor — Bosch BME688 (or BME690)

Function

A 4-in-1 environmental + gas sensor that detects VOCs, volatile sulfur compounds, CO, and hydrogen in the parts-per-billion range, plus temperature, humidity, and barometric pressure — all in one chip the size of a grain of rice. When driven with a programmable heater profile, it produces multi-channel gas resistance data suitable for pattern-recognition ML.

Specifications

Package	3.0 × 3.0 × 0.93 mm LGA-8
Weight	~1.5 mg
Gases detected	VOCs, VSCs, CO, H ₂ , and many others in ppb range
Environmental	T: −40 to +85 °C (±1 °C) · RH: 0–100 % (±3 %) · P: 300–1100 hPa (±1 hPa)
Interface	I ² C (up to 3.4 MHz) or SPI (up to 10 MHz)
Supply	1.71–3.6 V
Power	0.9 mA during gas scan · 3.7 µA idle at 1 Hz T/RH/P
I ² C addresses	0x76 or 0x77 (two per bus; more via I ² C multiplexer)
Chip price (volume)	~\$8.65 single, ~\$5 at 1k+
Source	bosch-sensortec.com — BME688

Recommended form factor

For the scent capture device, do NOT use individual bare chips — use the Bosch 8× shuttle board, which comes factory-populated with 8 identical sensors on a ~40 × 30 mm PCB with an edge connector. Running 8 sensors in parallel gives you massive statistical advantages: you can either average across 8 copies for noise reduction, or configure each sensor with a different heater profile for up to 80 virtual channels.

Note on the BME690

The BME690 is the direct successor to the BME688, released in 2025 with improved robustness in high-condensation environments. Pin-compatible drop-in replacement. Same software tooling. For any new design, spec the BME690 instead of the BME688. Both the 8× shuttle board and the Bosch Development Kit are now available in BME690 variants.

4.2 Flow Verifier — Honeywell HAF or Sensirion SFM3019

Function

Same role as in the smell.Board version: confirm that every capture saw the same volume of air. This is what lets you compare two captures a week apart and trust the comparison. Honeywell HAF at ~\$80 or Sensirion SFM3019 at ~\$60 are both proven options; either is fine.

For a low-cost prototype you can skip this part and rely on a tightly-controlled pump PWM duty cycle. For a shipping product, include it.

4.3 Suction Pump — KNF NMP 03

Unchanged from the smell.Board version. The pump is independent of the sensor choice — it's what pulls air reliably through the chamber. 24.2 × 12.9 × 21.4 mm, 0.33 L/min, 600 mbar vacuum, ~\$200 sample / less at volume. See original tech specs doc Section 3.3 for full details.

4.4 Intake Vent — GORE PolyVent Snap-In

Unchanged from the original spec. ePTFE hydrophobic membrane, ~17 mm OD, ~\$3 per unit at MOQ. See original tech specs doc Section 3.5.

4.5 Host Compute — Raspberry Pi CM4 (or ESP32-S3)

Primary recommendation

Raspberry Pi CM4 remains the best host for the first generation of the device. It runs full Linux, has Wi-Fi and Bluetooth on-module, has enough horsepower to run BSEC2 (Bosch's signal processing library) and any custom ML locally, and gives you a clean path to sync captures to your scent library over the network.

Alternative: ESP32-S3 module for volume / miniaturization

Because the BME688 is specifically designed for embedded microcontroller use, you can skip the CM4 entirely for a slimmer, cheaper, lower-power build. Bosch provides BSEC2 as a C library that runs on Cortex-M class microcontrollers. An ESP32-S3 module is ~18 × 25 × 3 mm, has built-in Wi-Fi + BLE, and costs \$3–5. This path would shrink the device to roughly 70 × 50 × 22 mm external — genuinely pocket-sized — at the cost of giving up Linux and the ability to run full-scale ML locally.

Recommendation: Start with the CM4 for the prototype so you have full flexibility. Migrate to ESP32-S3 once the capture cycle and ML pipeline are stable, if miniaturization matters for your product direction.

5. Physical Layout

The BME688 sensors are tiny (3 × 3 × 0.93 mm each) and the shuttle board that carries them is thin (~1.5 mm), so the entire sensing subsystem takes far less volume than the smell.Board. The device can be noticeably smaller.

Internal Layout (BME688 Edition) — Top View, to Scale

Scale: 1 mm ≈ 9 px · Interior: ~95 × 65 mm · Minimum height: ~25 mm · Smaller than smell.Board version

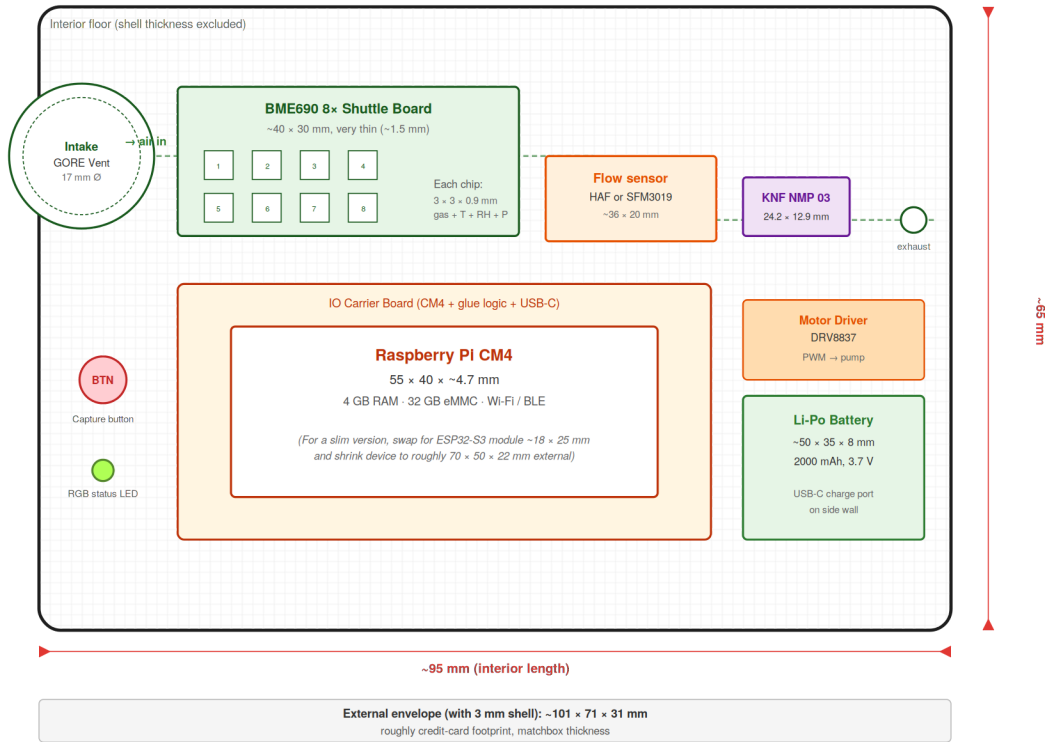


Figure 2 — Top-view internal layout, drawn to scale. The BME690 8× shuttle board sits in the airflow path; the CM4 carrier board sits underneath, outside the air channel.

5.1 Size budget

Component	L (mm)	W (mm)	H (mm)
BME690 8× shuttle board	~40	~30	~1.5
KNF NMP 03 pump	24.2	12.9	21.4
Flow sensor (HAF)	36	20	10
GORE PolyVent	17 Ø	17 Ø	12.9
Raspberry Pi CM4	55	40	4.7
Li-Po battery (2000 mAh)	~50	~35	~8
Minimum internal envelope	~95	~65	~25
External (3 mm shell, CM4 build)	~101	~71	~31
External (ESP32-S3 build)	~70	~50	~22

The CM4 build is about 15 % smaller than the smell.Board version. The ESP32-S3 build is genuinely pocket-sized, roughly the size of a Zippo lighter plus a small battery pack.

6. Electrical Connections

Much simpler than the smell.Board version — the entire sensor array communicates over a single I²C bus. Eight BME688 chips share the bus through a single I²C multiplexer (TCA9548A, ~\$2) because each chip only has two address options (0x76 and 0x77) and we need eight.

Connection	Protocol	Pins / address	Purpose
BME688 ×8 → TCA9548A multiplexer	I ² C	8 channels, each a BME688 at 0x76	Isolate the 8 sensors
TCA9548A → CM4	I ² C	I ² C-1, mux at 0x70	Single bus to host
Flow sensor → CM4	I ² C	I ² C-1, separate address	Flow verification
Pump motor → driver → CM4	GPIO / PWM	GPIO18 (PWM0) + GPIO17 (EN)	Speed + on/off
Capture button + LED	GPIO	Any free GPIOs	User I/O
Battery → PMIC → CM4	Power rail	5 V input on carrier	System power

Note: the BME688 natively has a T/RH/P sensor inside it, so the separate SHT45 from the original spec is redundant here. You get environmental correction data for free on every gas reading from every chip.

7. Capture Cycle (Software Behavior)

The capture cycle differs slightly from the smell.Board version because the BME688 has its own built-in heater scheduler. Instead of streaming raw resistance continuously, you configure a heater profile and the sensor returns one gas reading per heater step.

- **1. Warm-up / purge** — pump runs at low PWM for ~5 s, sensors complete 2–3 heater cycles to stabilize. Baseline 80-dim vector recorded.
- **2. Trigger** — user presses capture button. LED turns amber.
- **3. Sample window** — pump ramps to target flow rate. Each sensor runs its 10-step heater profile twice (~5 s). Host logs: 8 × 10 = 80 gas resistance values per cycle, T/RH/P from each chip, flow reading continuously.
- **4. Recovery / purge** — pump runs at higher flow for ~3 s to clear sensors. LED turns green when stable.

- **5. Profile write** — host subtracts baseline, normalizes by T/RH and flow, computes fingerprint vector, writes to local storage, queues for cloud sync.

7.1 Scent profile object

Field	Description
capture_id	UUID assigned at capture time
timestamp	ISO 8601 UTC
user_label	User-supplied scent name
heater_profile	Array of 10 target temperatures (°C) used this capture
raw_gas_matrix	8 sensors × 10 heater steps × N cycles, gas resistance in ohms
baseline_matrix	Same shape as raw_gas_matrix, captured during warm-up
env_log	Time-series of T, RH, P from each chip (8 × time)
flow_log	Time-series of flow rate in SCCM
fingerprint_vector	80-dim normalized response (sample – baseline, env-corrected)
fingerprint_hash	Hash of quantized fingerprint; used as library key
device_meta	Firmware version, shuttle board serial, calibration date

8. Bill of Materials

#	Part	Role	Interface	Est. price
1	Bosch BME690 8× Shuttle Board	Gas sensor array	I ² C	\$25
2	TCA9548A I ² C multiplexer breakout	Bus expander	I ² C	~\$7
3	KNF NMP 03 (KPDC-L)	Suction pump	PWM motor	~\$200 sample
4	Honeywell HAF flow sensor (optional)	Flow verification	I ² C	~\$80
5	GORE PolyVent Snap-In	Intake barrier	mechanical	~\$3
6	Raspberry Pi CM4 (4GB, 32GB eMMC, Wi-Fi)	Host compute	n/a (host)	~\$80
—	Carrier PCB + motor driver + battery + USB-C + shell	Support	—	~\$40

	Prototype BOM total (CM4 build, all-in)			\$235–\$280
	Alt ESP32-S3 build total			\$160–\$200

Comparison: the smell.Board edition's prototype BOM came in at \$830–\$920. This edition is 65–75 % cheaper per unit. At production volume, the gap widens further — the BME690 drops to ~\$3 at 10k units, while individual chip pricing from SmartNanotubes remains unclear.

9. Head-to-Head Comparison

Direct comparison between the two sensor choices, independent of the rest of the stack.

Attribute	smell.Board (Edition A)	BME690 8× shuttle (Edition B)
Dev kit price	€389 (\$430)	~\$25 shuttle + ~\$90 app board
Chip price at volume	Unknown, not publicly listed	~\$3–\$5 at 1k+ units
Sensor technology	Carbon nanotube chemiresistors	MEMS metal-oxide hot plate
Distinct chemistries	15–16 different nanotube coatings	1 metal oxide, differentiated by heater temperature
Fingerprint dimensionality	~15 features (manual confirms)	Up to 80 (8 chips × 10 heater steps)
T/RH correction	External (SHT45 needed)	Built into every sensor chip
ML tooling	Smell Annotator (cloud-tied, login required)	BME AI-Studio (local, no login)
Data ownership	Raw data local, but models trained in SmartNanotubes cloud	Everything local, your infrastructure
Ecosystem / community	Small, vendor-specific	Massive (Arduino, ESP32, RPi, Adafruit, SparkFun)
Vendor risk	Series-A startup (~€2.4M raised)	Bosch (Fortune Global 100, millions of sensors/yr)
Best-in-class at	Discriminating complex organic smells (coffee, food aromas)	VSCs (breath, spoilage), air quality, industrial VOCs

Unknown (needs testing)	How well it holds up on novel smells it wasn't trained for	How well it discriminates fragrance-grade complex smells
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10. Comparison Test Protocol

Before committing to either sensor for the production product, run both in parallel against a reference smell set. This section specifies the exact protocol so results are repeatable and defensible.

10.1 Objective

Determine which sensor can reliably distinguish between 10 reference smells of varying similarity, with enough repeatability that a smell captured today produces a fingerprint close to the same smell captured a week later on a different device unit.

10.2 Reference smell set

Use the following 10 reference smells. They're chosen to span easy, medium, and hard discrimination cases — a good sensor should nail the easy ones and give you a realistic picture of where the hard boundaries sit.

#	Smell	Source	Difficulty
1	Freshly ground coffee	~5 g beans, ground	Easy (Bosch ships a demo for this)
2	Vanilla extract	3 drops on a cotton pad	Easy
3	Lemon zest	Freshly grated peel	Easy
4	Isopropyl alcohol	70 % IPA on cotton	Easy (strong, clean signal)
5	Rosemary	Fresh sprig, crushed	Medium
6	Leather	Wallet or belt, enclosed jar	Medium
7	Cedar wood	Shavings or block	Medium
8	Rose (essential oil)	1 drop diluted 1:10 in carrier oil	Hard (complex floral)
9	Jasmine (essential oil)	1 drop diluted 1:10	Hard (tests floral discrimination vs. rose)
10	Baseline (clean room air)	Unscented test chamber	Control

10.3 Experimental setup

- Use the same sealed 500 mL glass jar for every smell. Place the smell source inside, cap the jar with a modified lid that has a 10 mm hole, and hold the sensor intake 2 cm above the hole for sampling.
- Run every capture in the same room at the same time of day (within a 2-hour window) to minimize ambient-air confounds.
- Record ambient T, RH, and outdoor weather for each session in a notes field — sensor drift correlates with these.
- Allow 10 minutes of clean-air recovery between smells to let sensors return to baseline.
- Capture each smell 5 times per device per day, across 3 days, with both sensors running simultaneously on the same air sample where possible. Total: 2 sensors × 10 smells × 5 captures × 3 days = 300 captures.

10.4 Metrics

Compute and report the following for each sensor:

- **Within-class cosine similarity:** mean similarity between the 15 captures of the same smell (across 3 days). Higher is better. A good sensor lands >0.90.
- **Between-class cosine similarity:** mean similarity between captures of DIFFERENT smells. Lower is better. A good sensor lands <0.70 on easy smells, <0.85 on hard ones like rose-vs-jasmine.
- **Discrimination margin:** (within-class similarity) – (between-class similarity), computed per pair. Margin >0.15 on the hard pairs is the bar for "ready for library use."
- **Day-to-day drift:** mean distance from day-1 centroid to day-3 centroid for the same smell. Lower is better; this tells you how often the library will need recalibration.
- **Classification accuracy:** train a simple classifier (k-nearest-neighbor or small MLP) on day-1 and day-2 captures, test on day-3. Report top-1 accuracy and confusion matrix. Target: >85 % top-1.

10.5 Decision rule

Outcome	Recommendation
BME690 hits >85 % classification AND >0.15 margin on hard pairs	Ship with BME690. Saves \$600+ per unit. No reason to pay the nanotube premium.
BME690 fails hard pairs (rose vs. jasmine) but nails easy ones	Depends on product. If library is curated (100 distinct smells, mostly food + natural), BME690 still wins. If users must distinguish hundreds of floral fragrances, smell.Board may be needed.
Both sensors fail repeatability (within-class <0.85)	Pause. The product concept itself may be harder than commercial sensors can deliver. Reconsider pairing with GC-MS for a subset of reference smells.
smell.Board clearly outperforms BME690	Use smell.Board for flagship product, but plan a BME690-based cheaper tier for mass market. The AI-Studio infrastructure still applies.

10.6 Time and cost estimate for the test

- Hardware: ~\$500 total (smell.Board kit already acquired + \$280 for BME690 build + \$100 for test supplies).
 - Bench time: ~40 hours across 3 weeks (includes firmware for both sensors, capture sessions, analysis).
 - Deliverable: comparison report with metrics, confusion matrices, and a clear recommendation.
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11. Open Questions & Assumptions

- **BME688 on fragrance-grade smells:** Bosch has publicly demonstrated coffee, VSC (spoilage), and air-quality applications. They have NOT publicly demonstrated perfume-grade floral discrimination. The comparison protocol in Section 10 addresses this directly. Do not commit to BME688 for a fragrance product without running the test.
 - **BSEC2 licensing:** Bosch's signal processing library (BSEC2) has a redistribution license that allows use in commercial products but requires agreeing to specific terms. Read the license before shipping. For the prototype, no licensing issue.
 - **Heater profile optimization:** The 10-step heater profile is user-configurable. The default profile is optimized for VSC detection. For scent capture, you'll want to experiment with profiles that spread temperatures more evenly (e.g., 100, 150, 200, 250, 300, 320, 340, 360, 380, 400 °C) to maximize distinct chemical responses. This is its own research task.
 - **Sensor lifetime under continuous use:** BME688 has a rated lifetime of several years in typical consumer use. For a device that might run hundreds of heater cycles per day, expect degradation over 1–2 years. Plan for sensor replaceability in the industrial design.
 - **Reproduction side:** Unchanged from Edition A. The sensor choice does NOT affect the fundamental difficulty of translating a fingerprint back into a physical vial. That remains a separate problem requiring either a library of reference aromachemicals or a GC-MS analysis step for each recreated smell.
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12. Next Steps

- Order the BME690 Development Kit (8× shuttle board + application board 3.1) from Mouser or Buyzero. Total ~\$115. Ships in a few days.
- Download BME AI-Studio from Bosch's website. It's free, runs on Windows, no account required.
- Run the Coffee demo from Bosch to sanity-check the tooling end to end. Takes half a day.
- Execute the comparison protocol in Section 10 over 2–3 weeks.
- Based on results, commit to one sensor for the production design. Start enclosure CAD and carrier PCB layout.
- In parallel, begin scoping the reproduction-side tech spec (separate document): library of reference aromachemicals, GC-MS partnership for unknown captures, or a blend-from-known-bases architecture.